Manifold Steering Reveals the Shared Geometry of Neural Network Representation and Behavior

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Background: what is activation steering?

- Inside a neural network, each input is represented at every layer by a vector of numbers, called an **activation**
- **Activation steering** changes this vector during the forward pass, in order to change the model's output without retraining it
- Standard approach (*linear steering*): add a fixed direction, a *steering vector*, to the activation, multiplied by a strength (scalar) value
- This is motivated by the **linear representation hypothesis**: each concept is stored along a single direction in activation space
- In practice, linear steering often gives poor results: less fluent text, less diverse outputs, and unintended side effects

The key idea of the paper

- Many concepts are stored on **curved** shapes, e.g., days of the week form a circle
- Such a smooth curve or surface inside activation space is called a **manifold**
- A straight line between two points on a curve leaves the curve, through regions the model normally never uses
- The paper compares three paths: linear, along the manifold, and derived from the outputs (pullback)

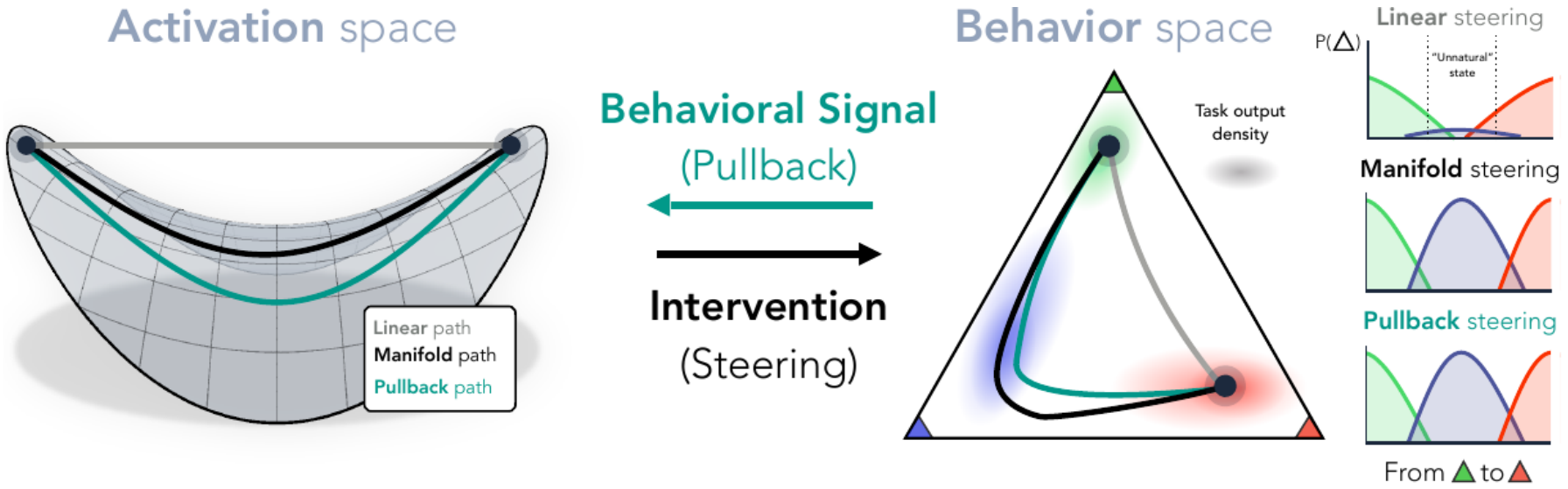


Figure 1. Left: paths in activation space. Right: the output distributions they produce, shown on a 3-concept probability triangle.

Hypothesis and approach

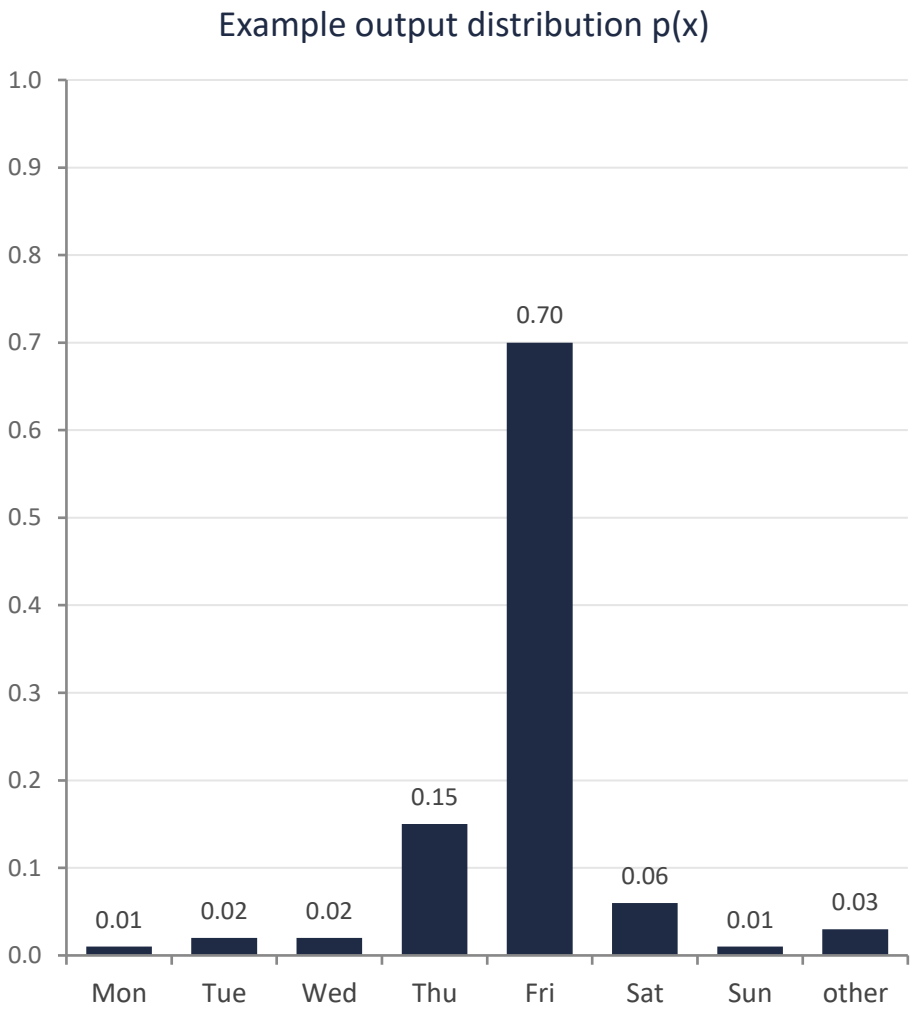
- For each task, the paper fits two manifolds:
  - the **activation manifold** $\mathcal{M}_h$: the shape formed by the model's internal activations
  - the **behavior manifold** $\mathcal{M}_y$: the shape formed by the model's output probability distributions
- **Hypothesis:** both are views of the same underlying concept structure, so their geometry should match
- Three tests of the link $\mathcal{M}_h \leftrightarrow \mathcal{M}_y$:
  - Test 1 (correlation): are distances along the two manifolds proportional?
  - Test 2 ($\mathcal{M}_h \rightarrow \mathcal{M}_y$): does steering along $\mathcal{M}_h$ produce outputs along $\mathcal{M}_y$?
  - Test 3 ($\mathcal{M}_h \leftarrow \mathcal{M}_y$): do activations chosen to produce outputs along $\mathcal{M}_y$ lie on $\mathcal{M}_h$?

Running example and tasks

- Running example: *What day is k days after z?* with $z \in \mathcal{Z} = \{\text{Mon}, \ldots, \text{Sun}\}$ and $k \in \{1, \ldots, 7\}$
- The concept has a natural structure: neighboring days are similar, and the week wraps around (a cycle)
- The model's answers reflect this: for *four days after Monday*, most probability goes to Friday and the rest mainly to Thursday and Saturday
- Four tasks, all run on Llama 3.1 8B:
  - **Cyclic:** weekdays and months
  - **Sequential:** letters and ages, e.g., *Alice is 7, Bob is 5 years older. How old is Bob?*

Two spaces: activations and outputs

- **Activation space** $\mathcal{A} = \mathbb{R}^n$: $h(x)$ is the model's activation vector for input x (layer 28, last token)
- **Behavior space** $\mathcal{Y} = \Delta^{|\mathcal{Z}|}$: all probability distributions over the concept values, plus one 'other' class for every remaining token
- $p(x) \in \mathcal{Y}$ is the model's output distribution for input x
  - Example (chart): for *four days after Monday*, 7 days + 'other' = 8 numbers that sum to 1, with most mass on Friday
- **Centroids:** activations and outputs are averaged over all prompts that have the same answer
  - e.g., *two days after Monday* and *three days after Sunday* both contribute to the Wednesday centroid



Fitting the activation manifold $\mathcal{M}_h$

- **Step 1:** compress each activation to 64 dimensions with PCA, which keeps the directions of largest variation
- **Step 2:** compute one centroid per concept value, e.g., one point for each day
- **Step 3:** give each centroid an intrinsic coordinate — its position in the order (letters, ages), or its angle around the loop (days, months):

$$\theta = \operatorname{atan2}(\mathrm{PC}_2, \mathrm{PC}_1)$$

  - Example: a centroid at $(\mathrm{PC}_1, \mathrm{PC}_2) = (0.50, 0.87)$ has angle $\theta = 60°$
- **Step 4:** draw a smooth curve (a cubic *spline*) through the centroids
  - The spline s maps a coordinate to a point on $\mathcal{M}_h$; it is used later to move along the manifold

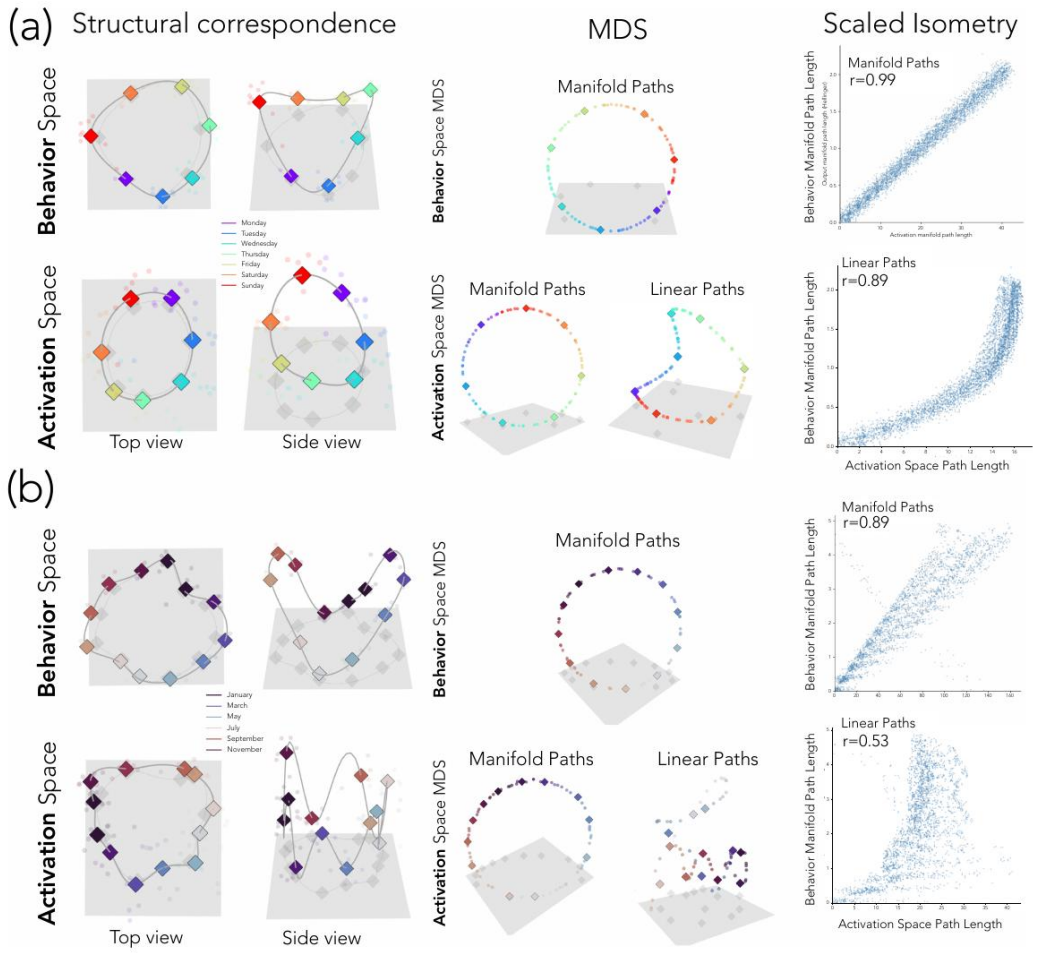


Figure 2: (a) weekdays, (b) months. Right column: distances along the two manifolds lie on a straight line.

Fitting the behavior manifold $\mathcal{M}_y$

- Probabilities cannot be compared well with ordinary straight-line distance, so each distribution is first transformed by taking square roots, $p \mapsto \sqrt{p}$
- Every distribution then lies on a sphere, and the **Hellinger distance** becomes an ordinary distance:

$$d_H(p, q) = 1/\sqrt{2} \parallel \sqrt{p} - \sqrt{q} \parallel$$

  - Example: $p = (0.7,0.2,0.1)$, $q = (0.2,0.7,0.1)$ give $\sqrt{p} \approx (0.84,0.45,0.32)$ and $\sqrt{q} \approx (0.45,0.84,0.32)$
  - Difference $(0.39, -0.39,0)$ has length 0.55, so $d_H \approx 0.55/1.41 \approx 0.39$
  - $d_H = 0$ for identical distributions and $d_H = 1$ for distributions with no overlap
- A spline is fitted through the square-rooted behavior centroids, kept on the sphere, and squared back into probabilities

Test 1: do the two manifolds have the same shape?

- Both manifolds show the concept's structure: days and months form **closed loops in the correct order**; letters and ages form open curves
- The loop in activation space was known before; the loop in output space is a new finding
- Test: compute the **geodesic distance** — the distance travelled along the manifold — between all pairs of points, in both spaces
- If the two sets of distances are proportional, the manifolds are related by a **scaled isometry**: the same shape at a different size
- Comparison: linear path distance in activation space

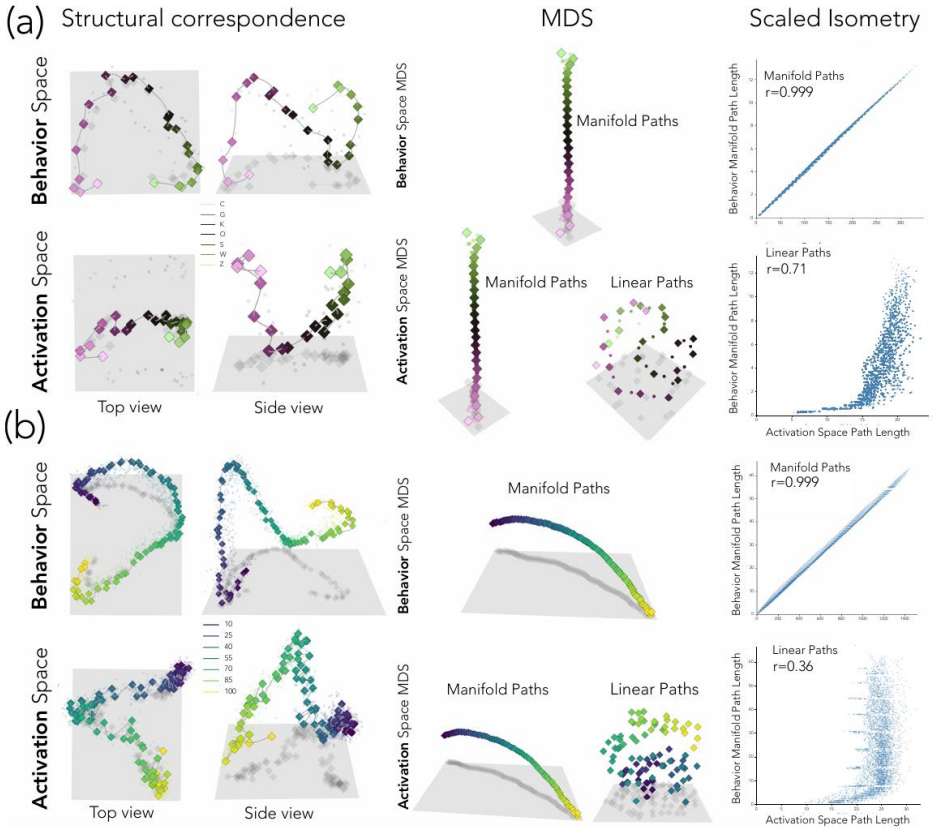
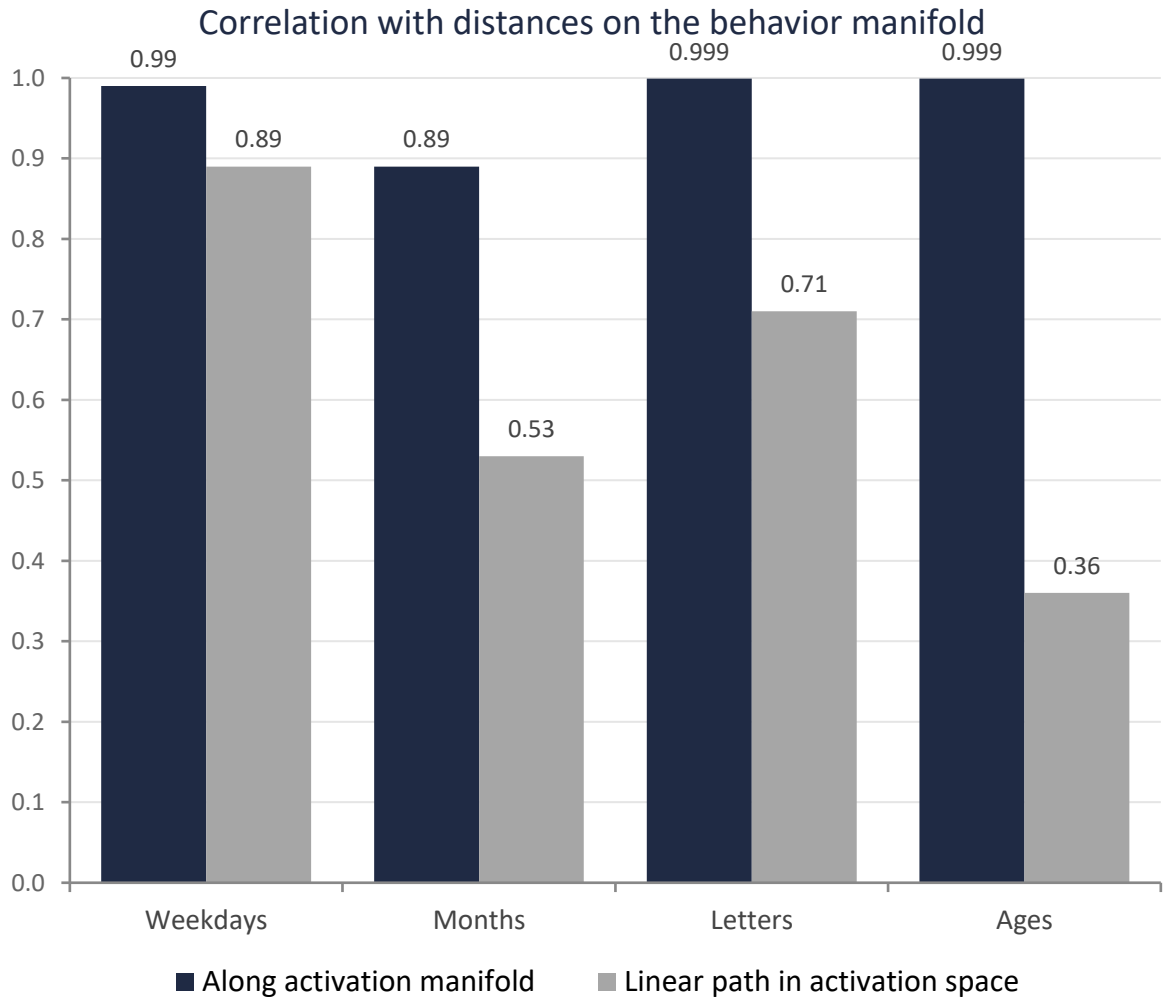


Figure 3: fitted curves for letters (a) and ages (b), in behavior and activation space.

Test 1 results: distances match along the manifolds

- The chart shows the correlation r between distances on $\mathcal{M}_y$ and distances in activation space, measured in two ways
- Distances **along the manifold** $\mathcal{M}_h$ match almost perfectly ($r \geq 0.89$ in every setting)
- **Linear path** distances match much less well, down to $r = 0.36$ for ages
- Conclusion: $\mathcal{M}_h$ and $\mathcal{M}_y$ are approximately a scaled isometry; linear path distance fits neither
- This suggests: moving along one manifold should correspond to moving along the other. Tests 2 and 3 check this by intervention



How an intervention works

- **Intervention:** replace the activation at the chosen layer with a target activation vector $h^\star$ and let the model finish the forward pass
  - The resulting output distribution is written $p_{h \leftarrow h^\star}(x)$
- **Steering path:** a curve π: $[0,1] \rightarrow \mathcal{A}$ from a start point ($t = 0$) to an end point ($t = 1$)
- Intervening at each point of the path gives a sequence of outputs $p_{h \leftarrow \pi(t)}(x)$, i.e., a path through behavior space
- **Question:** which path should we take between two points, e.g., from the *Tuesday* centroid to the *Friday* centroid?

Linear steering vs. manifold steering

- **Linear steering** moves along a straight line in activation space:

$$\pi_{\text{lin}}(t) = (1 - t)\, h_0^\star + t\, h_1^\star \qquad (1)$$

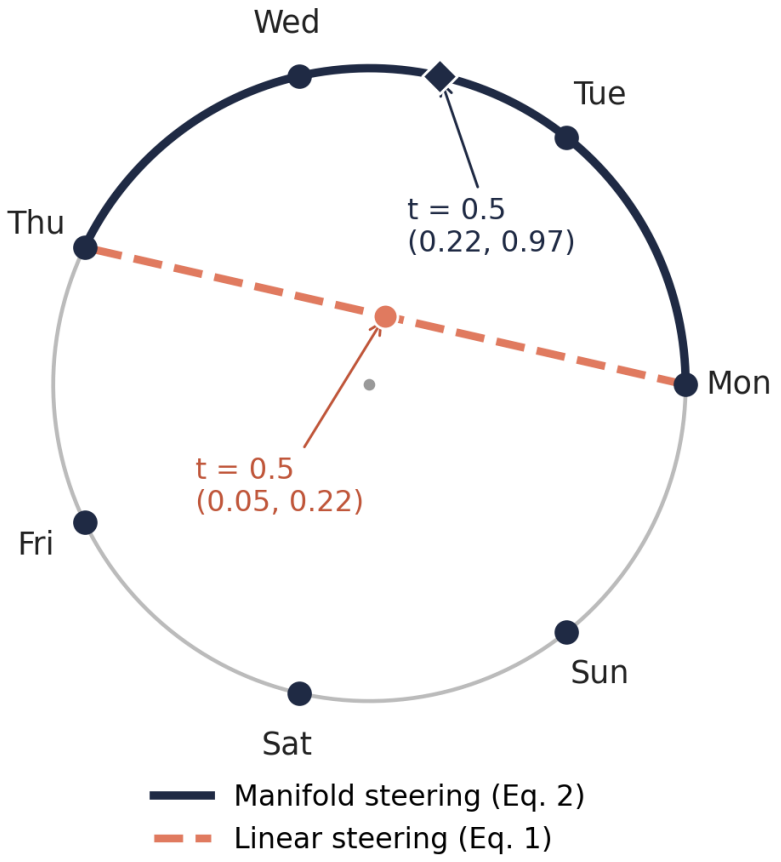
- **Manifold steering** moves evenly in the manifold's own coordinates u, then maps back to activations with the spline s:

$$\pi_{\text{m}}(t) = s\big((1 - t)\, u_0 + t\, u_1\big), \quad u_i = s^{-1}(h_i^\star) \qquad (2)$$

  - u_i is the intrinsic coordinate of an endpoint, e.g., its angle on the loop
- The manifold path always stays on $\mathcal{M}_h$; the straight line usually leaves it
- Setup: 50 points along each path; results averaged over 16 prompts

Worked example: Equations (1) and (2) on a circle

- Assume the days lie on a unit circle, $51.4°$ apart: Monday at $0°$, Thursday at $154°$
- Halfway along the path, $t = 0.5$:
  - **Eq. (1):** $1/2\,[(1,0) + (-0.90,0.43)] = (0.05,0.22)$ — only 0.22 from the centre, far from every day
  - **Eq. (2):** angle $1/2\,(0° + 154°) = 77°$, i.e., the point $(0.22,0.97)$ on the circle, between Tuesday ($51°$) and Wednesday ($103°$)
- Expected behavior: manifold steering passes through Tuesday and Wednesday; linear steering passes through an empty region the model does not normally use



Test 2 results: smooth transitions vs. 'teleportation'

- **Manifold steering** (bottom rows): probability moves step by step through neighboring values, e.g., Tue → Wed → Thu → Fri
- **Linear steering:** probability jumps from the start value to the end value, skipping the values in between (*teleportation*)
  - Near the middle, unrelated 'other' tokens can receive more probability than any day
- In behavior space (top row), manifold-steering outputs follow $\mathcal{M}_y$; linear-steering outputs leave it

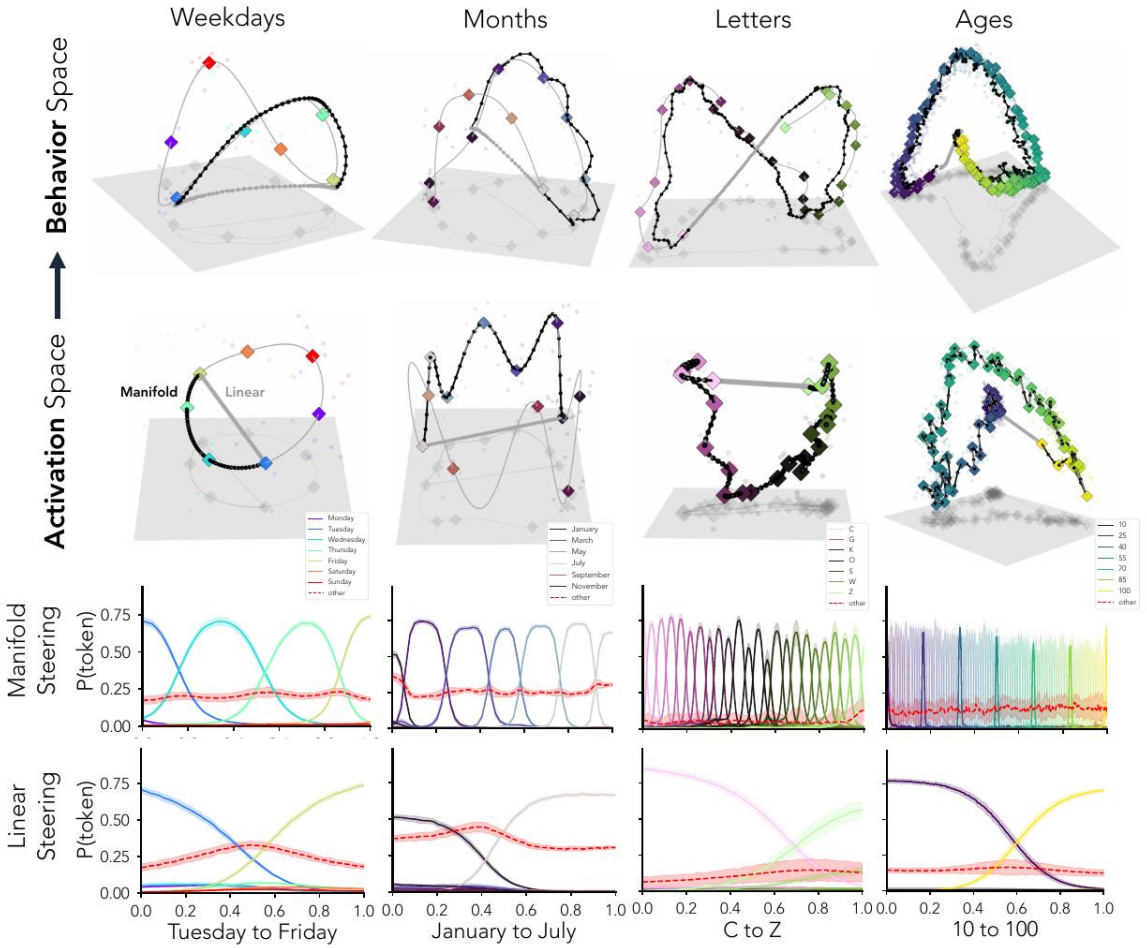


Figure 4. Rows: behavior space, activation space, P(token) under manifold steering, P(token) under linear steering.

Measuring how natural the outputs are

- Outputs of the unmodified model lie on or near $\mathcal{M}_y$; an output far from $\mathcal{M}_y$ is *unnatural*
- The paper adds up the distance to $\mathcal{M}_y$ along the whole output path $\gamma(t)$, called the **energy**:

$$E_{\mathrm{BC}}(\gamma) = \int_0^1 d_{\mathrm{BC}}\left(\gamma(t), \mathcal{M}_y\right) dt \qquad (3)$$

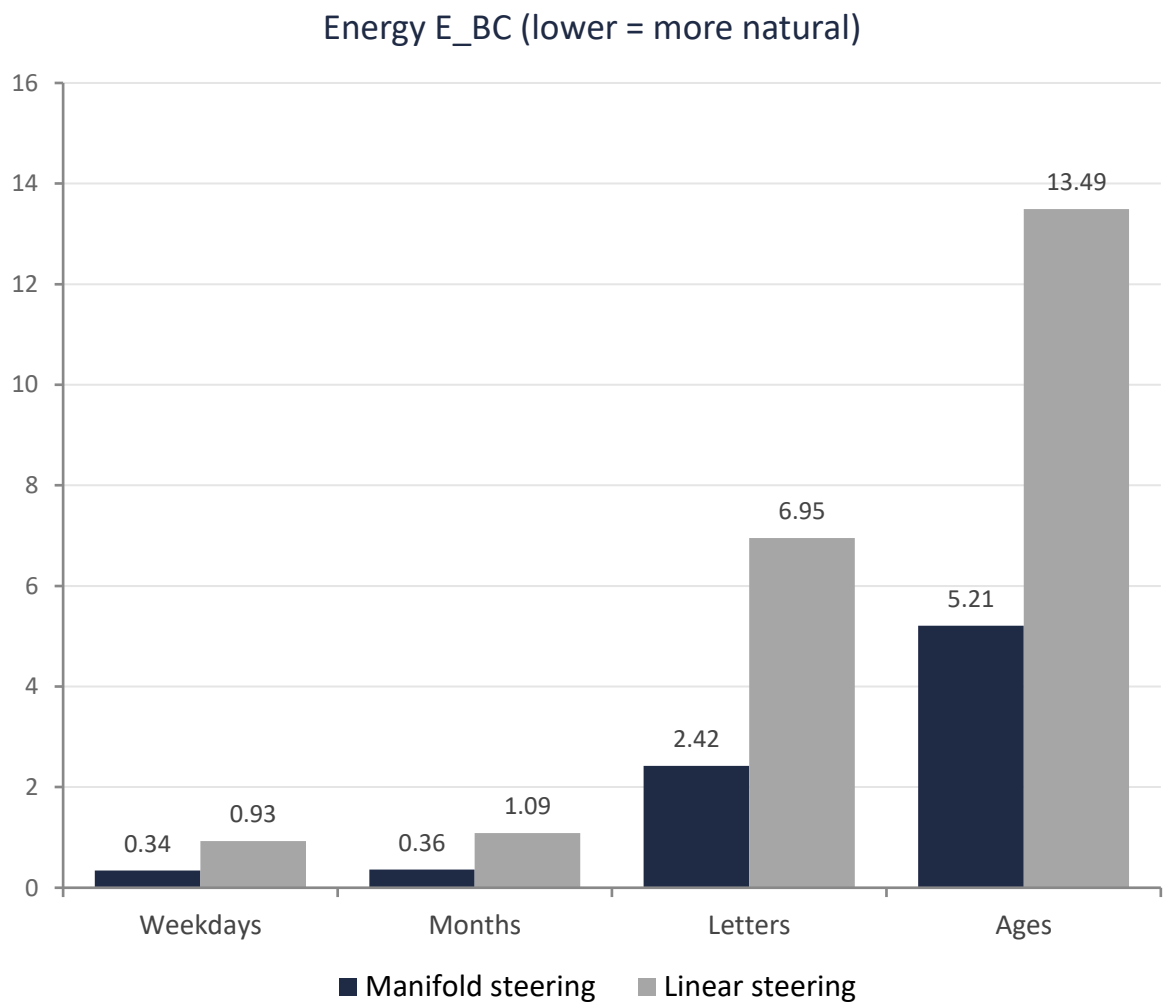
- d_{BC} is the Bhattacharyya distance to the nearest point q on $\mathcal{M}_y$:

$$d_{\mathrm{BC}}\left(p, \mathcal{M}_y\right) = \inf_{\sqrt{q} \in \mathcal{M}_y} -\log \sum_i \sqrt{p_i q_i}$$

  - Example: $p = (0.7, 0.2, 0.1)$, $q = (0.2, 0.7, 0.1)$: $\sum_i \sqrt{p_i q_i} = 0.37 + 0.37 + 0.10 = 0.85$, so $d_{\mathrm{BC}} = -\log 0.85 \approx 0.16$
  - If $p = q$, the sum is 1 and $d_{\mathrm{BC}} = 0$. Lower energy means a more natural output path

Test 2 results: manifold steering is more natural

- Manifold steering has much lower energy than linear steering on all four tasks
- Ratios (linear ÷ manifold): weekdays $0.93/0.34 \approx 2.7$, months 3.0, letters 2.9, ages 2.6
  - On average about **2.8×** lower energy; every difference is statistically significant ($p < 0.001$)
- Values differ between tasks because the tasks have different numbers of values and path lengths
- Conclusion: steering along $\mathcal{M}_h$ keeps the outputs close to the model's natural outputs ($\mathcal{M}_h \to \mathcal{M}_y$)



Test 3: from outputs back to activations (pullback)

- Reverse direction: start from a path of **natural outputs** along $\mathcal{M}_y$, then search for activations that produce it
- An optimizer (L-BFGS) adjusts a path in activation space so that intervening at each point gives the target output
- The search starts from the straight line and is never told about $\mathcal{M}_h$; only 32 of the 64 PCA dimensions may change
- The resulting path is called the **pullback**
- **Result:** pullback paths (teal) bend away from the straight line and follow $\mathcal{M}_h$, like the manifold-steering paths (black)

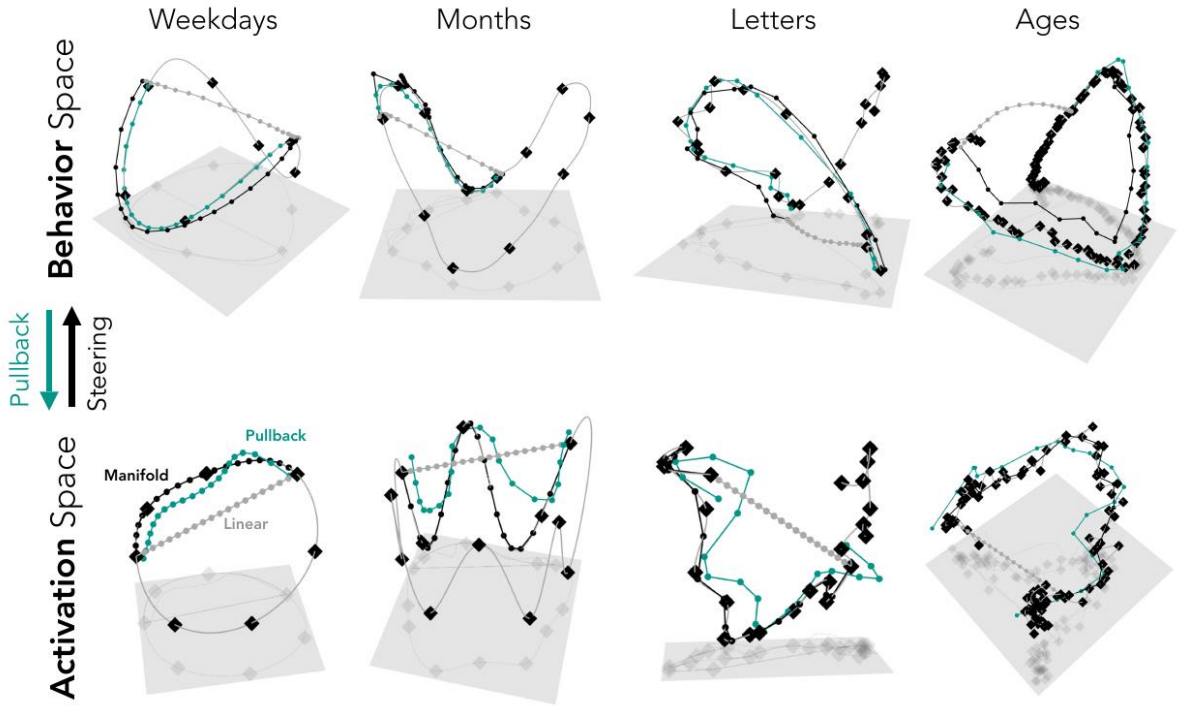


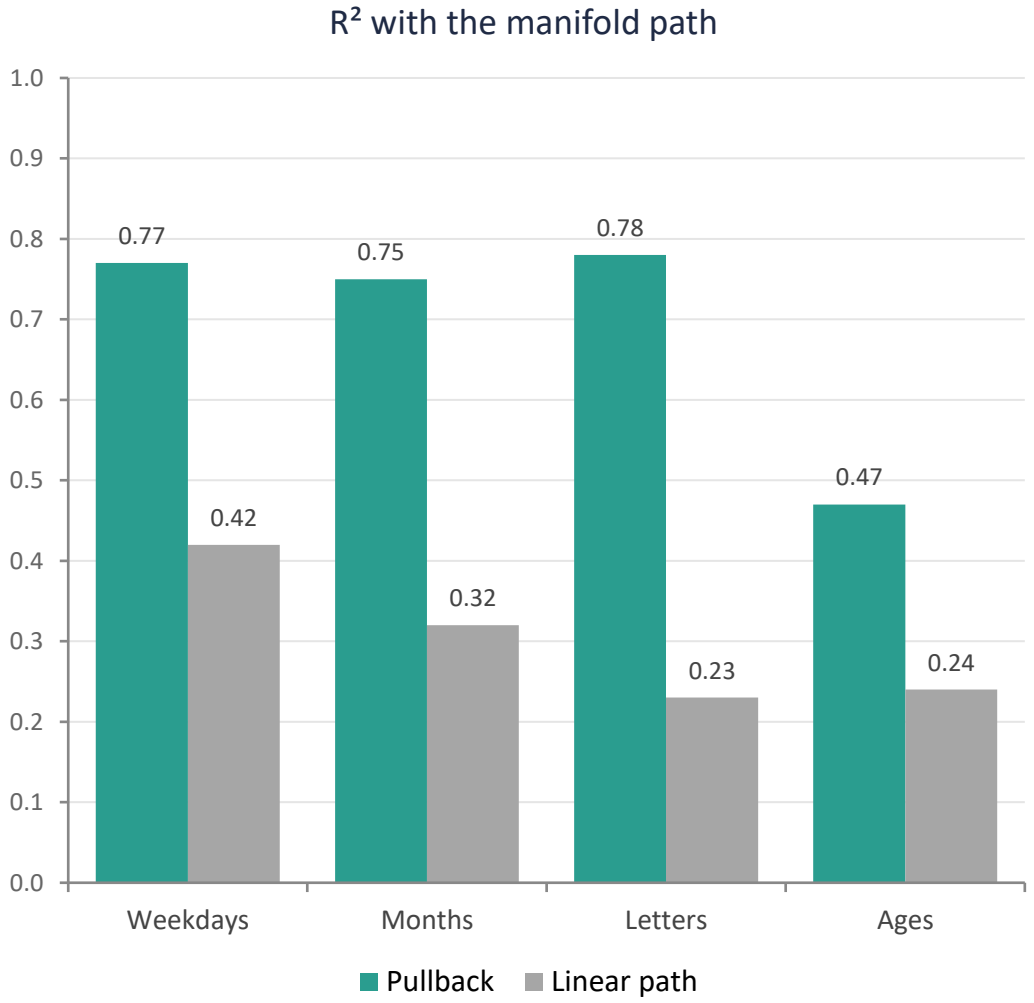
Figure 5. Top: behavior space. Bottom: activation space. Black = manifold steering, teal = pullback, grey = linear.

Test 3 results: how close is the pullback to $\mathcal{M}_h$?

- The paper measures this with an R^2 score (App. A.9):

$$R^2 = 1 - \frac{\sum_t \| \pi_h^{\mathrm{pb}}(t) - \mathrm{proj}_{\pi_h^*} \pi_h^{\mathrm{pb}}(t) \|^2}{\sum_t \| \pi_h^{\mathrm{pb}}(t) - \bar{\pi}_h^{\mathrm{pb}} \|^2}$$

  - Top: squared distances from pullback points to the manifold path π_h^*. Bottom: total variance of the pullback path
  - Example: distances summing to 2 and a variance of 10 give $R^2 = 1 - 2/10 = 0.8$; $R^2 = 1$ means a perfect match
- Pullback scores well above the linear path baseline on all tasks ($p < 0.001$)
- Two different sources — activations and outputs — point to the same paths



Theory: steering as choosing a geometry

- A **Riemannian metric** G is a rule for measuring length that may change from place to place: $G(h)$ tells how costly it is to move near the point h
- Length of a path π under G ($\dot{\pi}$ is the velocity along the path):

$$L_G(\pi) = \int_0^1 \sqrt{\dot{\pi}(t)^\top \, G\big(\pi(t)\big)\, \dot{\pi}(t)} \; dt \qquad (4)$$

- A **geodesic** is the shortest path under G. Each choice of G defines a steering method
- Example: the path $\pi(t) = (t, 0)$ has velocity $\dot{\pi} = (1,0)$
  - With $G = I$: $\sqrt{1} = 1$, so $L = 1$ (ordinary length)
  - With $G = 4I$: $\sqrt{4} = 2$, so $L = 2$. Paths avoid such expensive regions

Three geometries for steering (Definition 1)

$$G_I = I_n \qquad (5)$$

- Every region and direction costs the same → straight lines → **linear steering**

$$G_E(h) = (\alpha\, e^{-E(h)} + \beta)^{-1} I_n \qquad (6)$$

- $E(h) \propto -\log p(h)$ is low where natural activations are common, so moving on $\mathcal{M}_h$ is cheap and moving off it is expensive → **manifold steering**
- Example ($\alpha = 99$, $\beta = 1$): on $\mathcal{M}_h$, $e^{-E} \approx 1$ gives $G_E = 1/100\, I$; far away, $e^{-E} \approx 0$ gives $G_E = I$, so moving off costs $\sqrt{100} = 10$ times more

$$G_F(h) = J_F(h)^\top\, g_y\big(F(h)\big)\, J_F(h) + \epsilon\, I_n \qquad (7)$$

- Measures a step in activations by how much the **output** changes (J_F = derivative of the map F from activations to outputs) → **pullback**

- **Key claim:** G_E (from activations) and G_F (from outputs) give nearly the same paths; G_I does not

Extension 1: 2-D concepts learned in context

- Task: words are placed on a grid; the prompt is a random walk over it (2048 words); the model predicts a neighboring word
- From the context alone, the model builds a 2-D map of the grid in its activations
- Tested on a 5×5 grid and a 9×9 cylinder (left and right edges joined); the manifolds are smooth 2-D surfaces
- Isometry holds again: $r = 0.99$ along the manifolds vs. 0.90 (grid) and 0.81 (cylinder) for linear paths

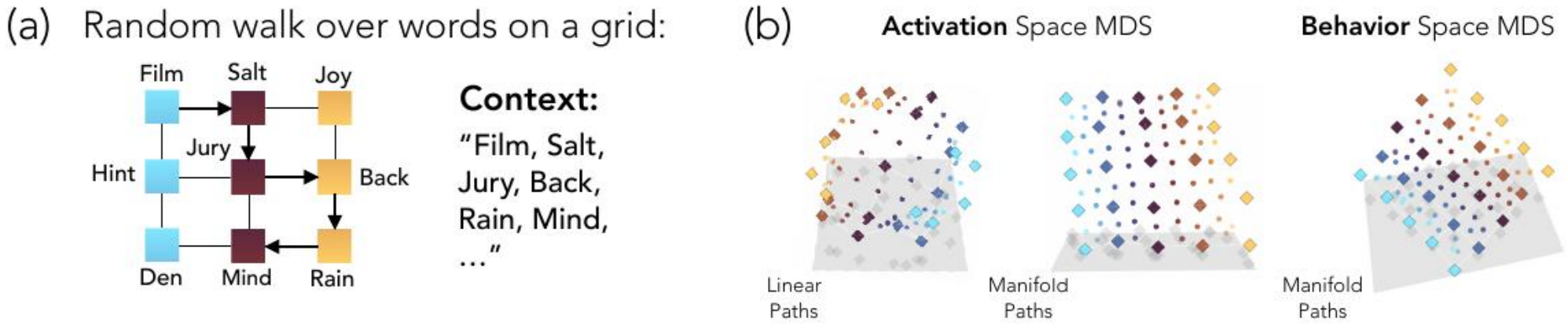


Figure 6(a, b). (a) The random-walk task. (b) Distances along both manifolds recover the grid; linear path distances distort it.

Extension 1 results: independent control of two directions

- Manifold steering along one coordinate moves the predicted position along one grid axis, while the other axis stays fixed
- The two manifold coordinates act as independent 'knobs': **factored control**
- Linear steering again jumps between the endpoints
- Same result on the 9×9 cylinder (App. C.1)

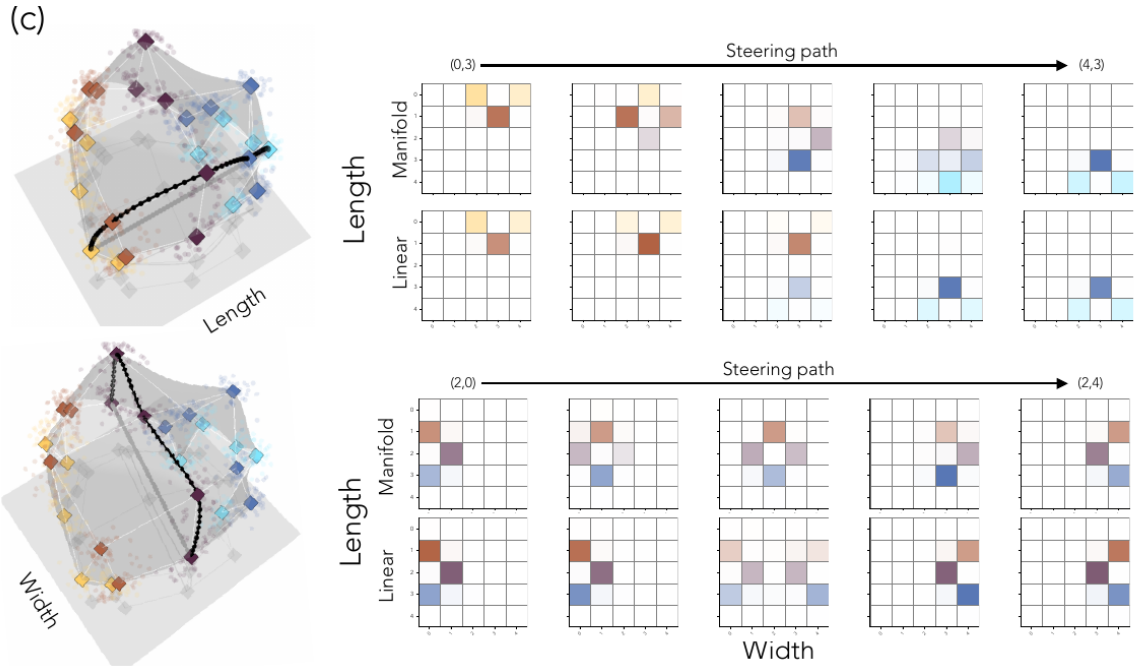


Figure 6(c). Each small grid shows where the model puts probability along the steering path.

Extension 2: a video world model (Mountain Car)

- **Mountain Car:** a car in a valley must build momentum to reach a goal; position $p \in [-1.2, 0.6]$
- A recurrent network predicts the next video frame from the current frame x_t and action a_t:

$$\mathbf{h}_t = \mathrm{GRU}([v_t;\, e(a_t)],\, \mathbf{h}_{t-1}), \quad v_t = \mathrm{LayerNorm}\big(f_{\mathrm{enc}}(x_t)\big) \qquad (8)$$

  - v_t: encoded image; $e(a_t)$: action embedding; $\mathbf{h}_t$: GRU memory ($n = 64$)
  - prediction $\hat{x}_{t+1} = x_t + f_{\mathrm{dec}}(\mathbf{h}_t)$
- Concept: the car's position; steering is applied to the image encoder's output

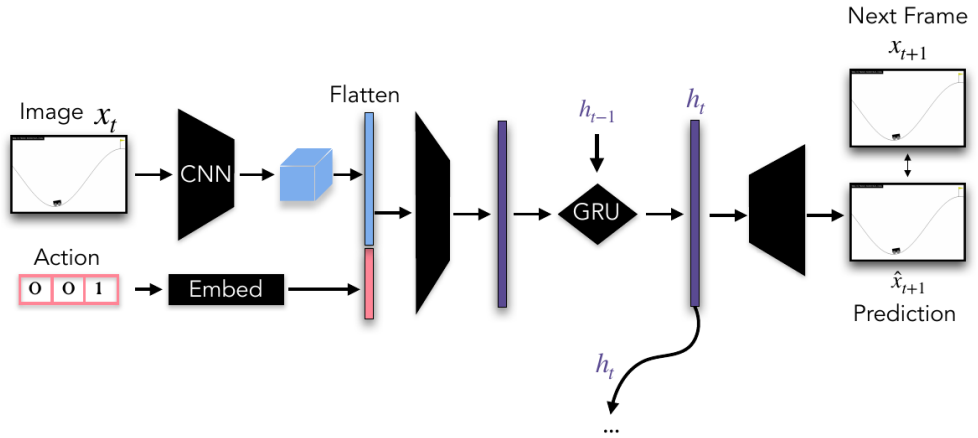


Figure 8. Encoder (CNN) → GRU → decoder.

World model: from activations to a belief about position

- Activation manifold: average the encoder activations in position bins and fit a smooth spline through the centroids μ_b
- The model's probability about position is computed from the distances of the activations to the centroids:

$$\mathrm{P(x)} = \mathrm{softmax}(-\frac{\| v - \mu_b \|_2}{\tau})_{b=1}^{B}, \qquad \tau = 0.5 \qquad (10)$$

  - Example: distances 0.5, 1.0, 1.5 → divide by τ and negate: −1, −2, −3 → softmax: (0.67, 0.24, 0.09). The closest bin gets most probability
- Behavior manifold: a spline through the probability bins, in square-root (Hellinger) coordinates
- Both manifolds are closed loops, because the encoder sees the wall ($p \approx -1.2$) and the goal ($p \approx 0.4$) as similar

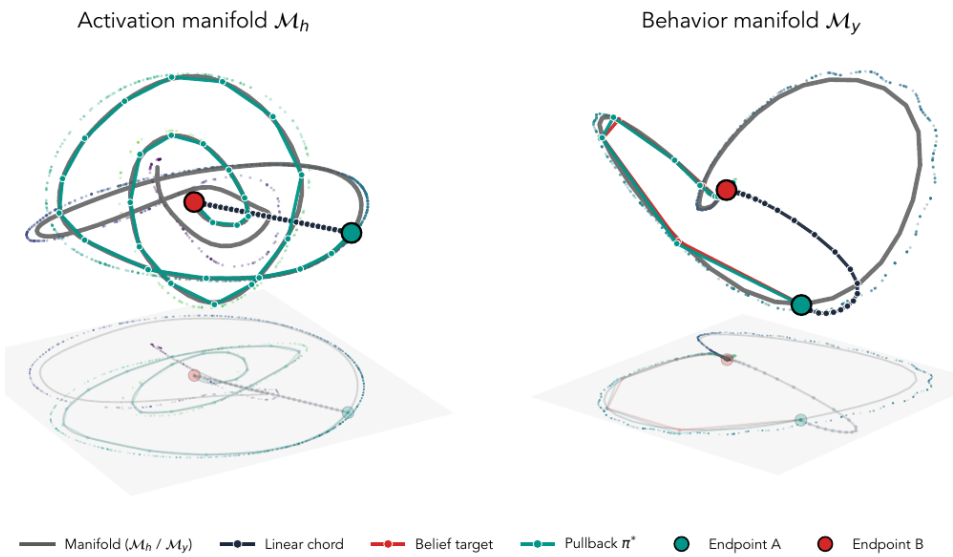


Figure 12: the activation manifold (left) and behavior manifold (right).

World model results

- **Manifold steering** (from $p = -0.4$ to $p = 0.4$): decoded frames show the car moving smoothly uphill
- **Linear steering:** frames show a blurred car in several places (mixed probabilities), then a jump to the end point
- **Isometry:** $r = 0.996$ along the manifolds, but $r = 0.06$ for linear path, because the activation curve folds back on itself
- **Pullback:** mean distance to $\mathcal{M}_h$ is 2.22 for the linear path, 0.29 for the pullback, 0.20 for the manifold path
  - Recovery: $(2.22 - 0.29)/(2.22 - 0.20) \approx 95\%$ of the way from the linear path to the manifold

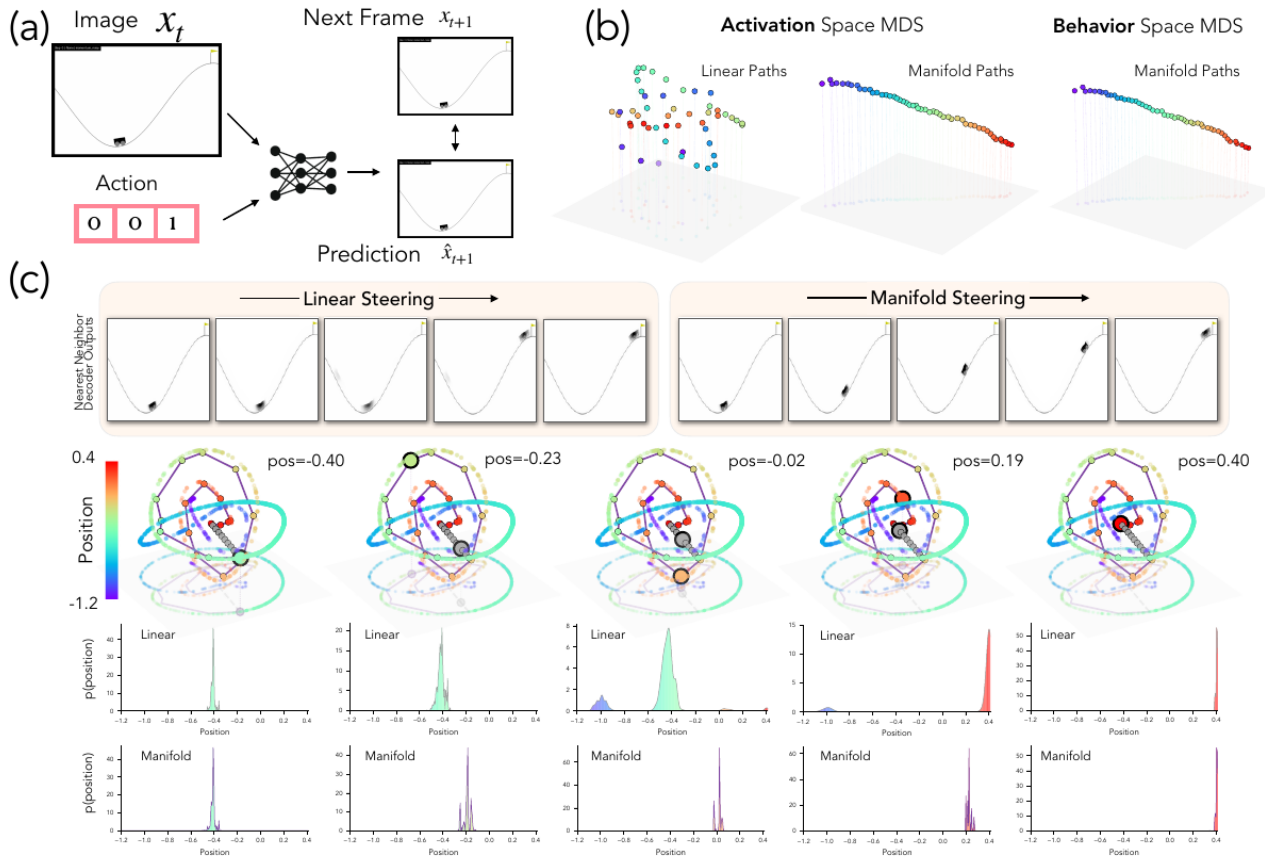


Figure 7. (a) Task. (b) MDS of distances. (c) Frames decoded along linear (left) and manifold (right) steering paths.

Discussion and limitations

- **Discussion**
  - Linear steering fails because it assumes flat geometry — not because steering activations is a bad idea
  - The shared geometry also appears for structures learned in context, so outputs are constrained by the activation geometry
  - Coordinates on fitted manifolds could serve as basic units for causal analysis of networks

- **Limitations**
  - Only simple concepts whose values are the outputs; abstract concepts (e.g., refusal, sycophancy) are untested
  - Only next-token outputs, not full generated sequences
  - Manifold fitting uses known labels and orderings; an unsupervised method is needed